

THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 6 – Options on Futures Contracts

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Outline

- What are futures options
- Put-call parity with futures options
- Bounds for futures options
- Comparing futures options with spot option prices
- Using binomial trees with futures options

Readings: Chapter 16

What are Futures Options?

- A futures option is the right, but not the obligation, to enter into a futures contract at a certain futures price by a certain date.
- The underlying security of a futures option is a futures contract, not a physical commodity. Exercising these options gives the holder a position in a futures contract.
- Futures options are generally American; that is, they can be exercised at any time during the life of the contract.

What are Futures Options?

A call futures option:

- is the right to enter into a long futures contract at a certain price; and,
- when the holder exercises the option they receive a cash amount equal to the excess of the futures price over the strike price.

A put futures option:

- is the right to enter into a short futures contract at a certain price; and,
- when the holder exercises the option they receive a cash amount equal to the excess of the strike price over the futures price.

What are Futures Options?

Example: You have bought a December call futures option on greasy wool with a strike price of $1,200 per bale.

- The asset underlying one contract is 100 bales of wool.
- As a result of purchasing this call futures option, you have the right to enter into a futures contract at the strike price, which in this case is $1,200 per bale.

We know that most futures options are American, which means you can exercise them before their expiry.

You decide to exercise your call futures option on greasy wool and the most recent settlement price for the December futures contract is $1,400. This means that the futures price when you exercise the call futures option is $1,400. Hence, F = $1,400.

What are Futures Options?

As a result of exercising this call futures option, you enter into a long futures position at $1,400.

- You also receive a cash amount (an increase in margin account) equal to the excess of the most recent settlement price for the December futures contract over the strike price:

($1,400-$1,200)x100 bales=$20,000. This follows the formula $(F-K)$

Now, when you decide to close out your December futures contract, the December futures price has risen to $1,500. The mechanics of your profit from closing out this futures position are identical to the profits from holding an ordinary futures contract.

- The gain on the futures contract is the excess of the December futures price at time of exercise over the price of the long futures contract you currently hold:

($1,500-$1,400)x100 bales=$10,000.

Thus you make a total gain of $30,000. $20,000 on the exercise of the option and the $10,000 on closing out the position.

Put-Call Parity

In lecture 4 we saw that put-call parity for a European call and put option on a non-dividend paying stock was:

$$c + Ke^{-rT} = p + S_0$$

To derive a put-call parity relationship for European futures options we consider the following two portfolios:

- Portfolio A: a European call futures option plus an amount of cash equal to Ke^{-rT}
- Portfolio B: a European put futures option plus a long futures contract plus an amount of cash equal to F_0e^{-rT}

Put-Call Parity

The cash in portfolio A is invested at the risk-free rate, and grows to K at time T.

Let F_T be the futures price at maturity of the option.

- If $F_T > K$, the call option in portfolio A is exercised and portfolio A is worth F_T.
- If $F_T < K$, the call is not exercised and portfolio A is worth K.

Therefore, the value of portfolio A at time T is therefore:

$$max(F_T, K)$$

The cash in portfolio B is invested at the risk-free rate, and grows to F_0 at time T.

The put option provides a payoff of $max(K - F_T, 0)$.

The futures contract provides a payoff of $F_T - F_0$.

The value of portfolio B at time T is therefore:

$$F_0 + (F_T - F_0) + max(K - F_T, 0) = max(F_T, K)$$

Put-Call Parity

Both portfolios have the same value at time T, namely:

$$max(F_T, K).$$

As they are European options and cannot be exercised early, they must have the same value today.

The value of portfolio A today is:

$$c + Ke^{-rT}$$

In portfolio B, the futures contract is worth zero today due to marking to market, so portfolio B is worth:

$$p + F_o e^{-rT}$$

Hence, put-call parity for a futures option is:

$$c + Ke^{-rT} = p + F_0 e^{-rT}$$

For American futures options, the put-call parity relation is:

$$F_0 e^{-rT} - K < C - P < F_0 - Ke^{-rT}$$ (end of chapter Q 16.19)

Put-Call Parity

Example:

- Imagine that the price of a European call option on greasy wool futures for delivery in six-months is $100 a bale, when the exercise price is $1,000.
- Assume that the greasy wool futures price for delivery in six months is currently $900, and the risk-free rate for an investment that matures in six months is 5% p.a.
- What should be the price of European put option on greasy wool futures with the same maturity and exercise date as the call option?

Put-Call Parity

Using:

$$c + Ke^{-rT} = p + F_0 e^{-rT}$$

We have:

$$100 + 1{,}000e^{-0.05x6/12} - 900e^{-0.05x6/12} = \$198$$

Therefore, the price of a European put futures option on greasy wool should be $198.

Lower Bounds for Futures Options

The put-call parity relationship we highlighted above, provides the lower bounds for European call and put options.

As the price of a put option cannot be negative, the lower bound for a European call must be:

$$c + Ke^{-rT} \geq F_0 e^{-rT}$$

OR $c \geq (F_0 - K)e^{-rT}$

Also, as the price of a call option cannot be negative, the lower bound for a put option is:

$$Ke^{-rT} \leq F_0 e^{-rT} + p$$

OR $p \geq (K - F_0)e^{-rT}$

Lower Bounds for Futures Options

As American options can be exercised at any time, the lower bounds for American options are:

$$C \geq F_0 - K$$

$$P \geq K - F_0$$

What happens if $C < F_0 - K$?

Comparing Futures Options with Spot Option Prices

The payoff from a European spot call option with strike price K is:

$$Max(S_T - K, 0)$$

The payoff from a European futures call option with the same strike price is:

$$Max(F_T - K, 0)$$

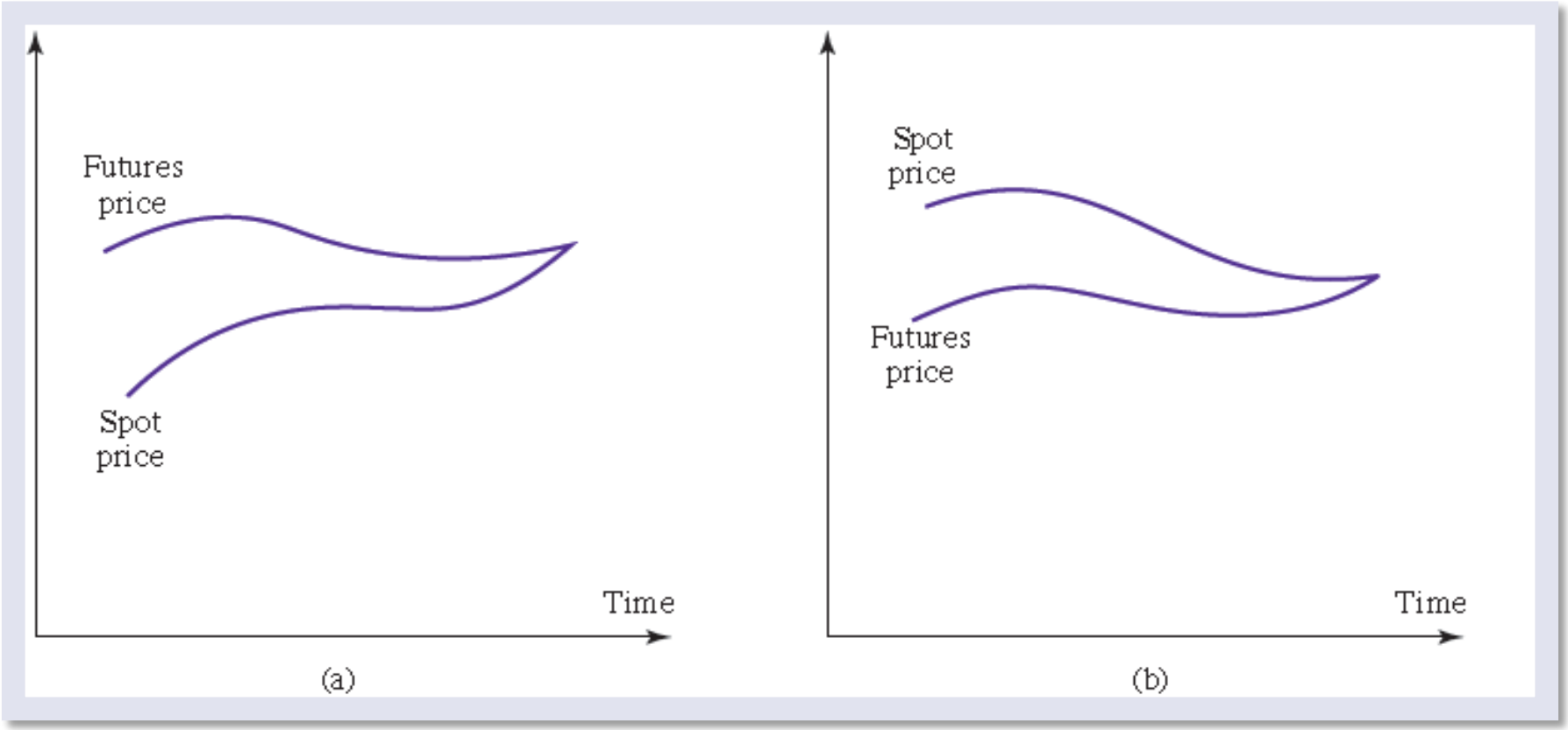
If the European call futures option matures at the same time as the futures contract, $F_T = S_T$ and the two options are equivalent.

If the European call futures option matures before the futures contract, it is worth more than the corresponding spot option in a normal market, and less in an inverted market.

Comparing Futures Options with Spot Option Prices

Normal market

Inverted market



Comparing Futures Options with Spot Option Prices

The payoff from a European spot put option with strike price K is:

$$Max(K-S_T,0)$$

The payoff from a European futures put option with the same strike price is:

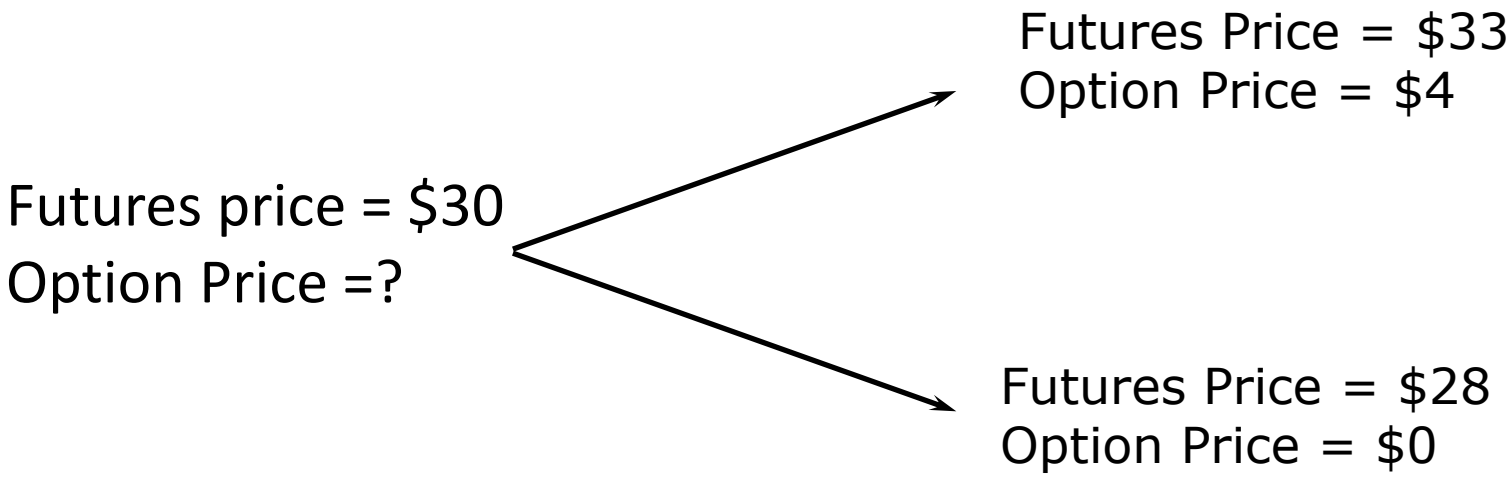
$$Max(K-F_T,0)$$

A European futures put option is worth the same as its spot option counterpart when the futures option matures at the same time as the futures contract.

If the European put futures option matures before the futures contract, it is worth less than the corresponding spot option in a normal market, and more in an inverted market.

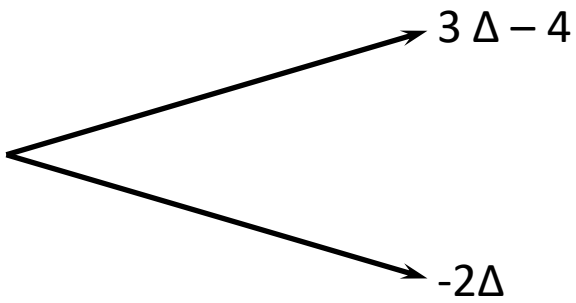
Valuation Using Binomial Trees

A one-month call option on futures has a strike price of 29.



Valuation Using Binomial Trees

- Consider a portfolio:
  - Short position in one options contract
  - Long position in Δ futures contracts



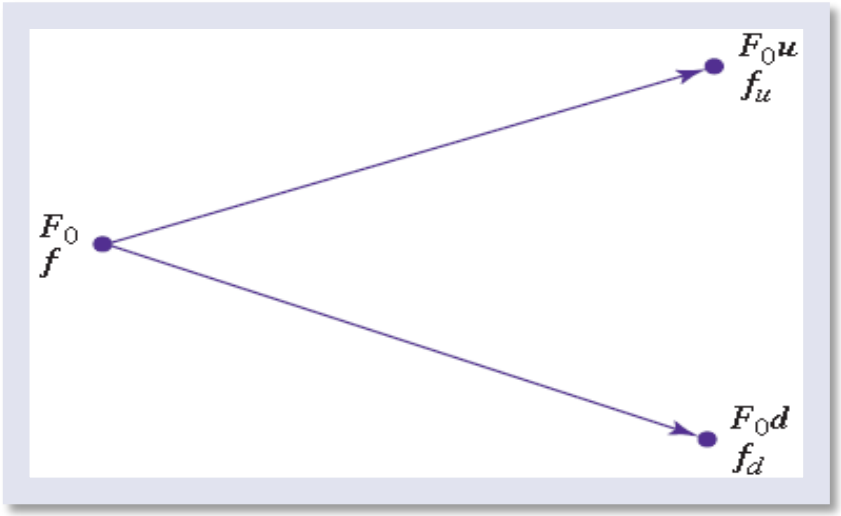
- Portfolio is riskless when $3\Delta - 4 = -2\Delta$ or $\Delta = 0.8$

What is the value of the portfolio today assuming r=.06?

- The value of the portfolio in one month = 3*0.8 – 4 = –1.6
- The value of the portfolio today = $-1.6e^{-0.06*1/12} = -1.592$
- Since the value of the futures contract today is zero, the value of the option today must be 1.592

Valuation Using Binomial Trees

Futures price and option price in the general situation



$$(F_0 u - F_0)\Delta - f_u$$

$$(F_0 d - F_0)\Delta - f_d$$

$$\Delta = \frac{f_u - f_d}{F_0 u - F_0 d}$$

Value of the portfolio at time T: $(F_0 u - F_0)\Delta - f_u$

Value of the portfolio today: $[(F_0 u - F_0)\Delta - f_u]e^{-rT}$

The present value of the portfolio: $-f = [(F_0 u - F_0)\Delta - f_u]e^{-rT}$

Valuation Using Binomial Trees

Substituting for Δ and simplifying reduces this equation to:

$$f = e^{-rT}[pf_u + (1-p)f_d]$$

Where $p = \frac{1-d}{u-d}$

Recall that for stock options

$$p = \frac{e^{rT} - d}{u - d}$$

Why?

Valuation Using Binomial Trees

We know that a futures contract requires zero investment.

We also know that in a risk-neutral world, the expected payoff from holding a position in an investment that requires zero initial investment must be zero.

The expected growth rate of the futures price in a risk-neutral world, must therefore also be zero.

Valuation Using Binomial Trees

Previously, we stated that a stock paying a dividend at rate q, has an expected growth rate of r-q.

If we set $q = r$, the expected growth rate of the stock price is zero, making it analogous to a futures price.

Therefore, futures prices behave in the same way as a stock paying a dividend yield at the risk-free rate.

Valuation Using Binomial Trees

We can therefore utilise the following result in lecture 7 to find the risk neutral probability distribution p for an option written on a futures contract:

$$pS_0u + (1-p)S_0d = S_0e^{(r-q)T}$$

Substituting r for q, and F_0 for S_0, yields the following risk neutral probability distribution p for an option written on a futures contract.

$$puF_0 + (1-p)dF_0 = F_0e^{(r-r)T}$$

solving for p

$$puF_0 + (1-p)dF_0 = F_0$$

$$p = \frac{1-d}{u-d}$$

Next Week

Black-Scholes-Merton option valuation model